Practical 6: Rendering External JSON Data using Fetch API, Search and Filter

Problem definition

Display event lists, student profiles, notices, or FAQs from external JSON files using Fetch API. Implement dynamic rendering, search, filter, sorting, and pagination.

Key questions

1. How is JSON fetched, parsed, and rendered?

JSON data is fetched from an external JSON file using the Fetch API.
The fetched response is parsed into JavaScript objects or arrays.
The parsed data is then processed using JavaScript.
Finally, the data is dynamically rendered and displayed on the HTML page.

2. Which array methods are used for search, filter, sort, and pagination?

The `filter()` method is used to search and filter the required data from an array.
The `includes()` method is used to match the search text with the available data.
The `sort()` method is used to arrange the data in ascending or descending order.
The `slice()` method is used to divide the data into smaller portions for pagination.

3. How are loading and error states handled?

A loading state is displayed while the JSON data is being fetched. After the data is successfully loaded, the loading message is replaced with the actual data. If an error occurs while fetching or processing the data, an error-handling mechanism displays an appropriate error message to the user.

4. How is modularity maintained in JavaScript files?

Modularity is maintained by dividing the JavaScript code into separate functions or modules. Each function or module performs a specific task such as fetching, searching, filtering, sorting, or rendering data. This makes the code more organized, reusable, readable, and easier to maintain.

Code:

Index.html

```
<!DOCTYPE html>
<html>
<head>
```

```
    <title>JSON Fetch Practical</title>
</head>

<body>
 <h1>Student Details</h1>
  <div id="student"></div>
 <script>
  // Fetch JSON file
  fetch("students.json")
  // Convert response into JSON
  .then(response => response.json())
  // Get JSON data
  .then(students => {

  let output = "";
  // Display each student
     students.forEach(student => {
  output += `
 <h2>${student.name}</h2>
   <p>Age: ${student.age}</p>
     <p>Course: ${student.course}</p>
   <p>Marks: ${student.marks}</p>
<hr>
    `;
  });

    // Display on webpage
     document.getElementById("student").innerHTML = output;
   })

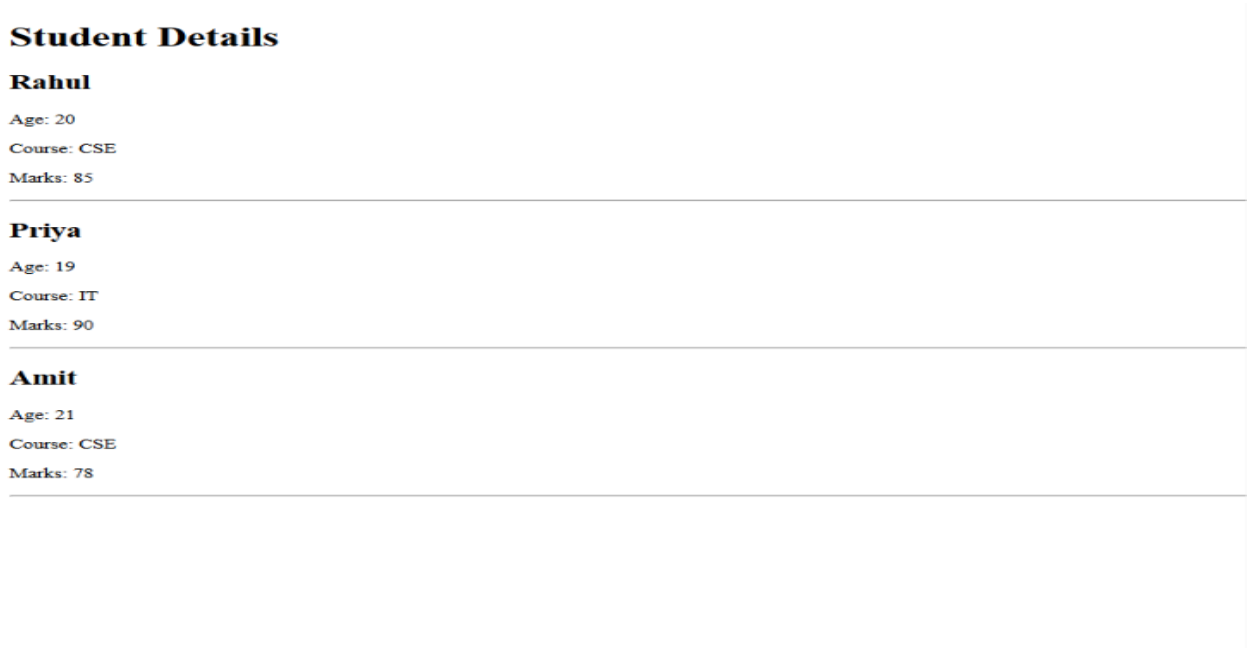
     // Error handling
  .catch(error => {
   console.log("Error:", error);
  });
 </script>

</body>
</html>
```

students.JSON

```
[
    {
        "name": "Rahul",
        "age": 20,
        "course": "CSE",
        "marks": 85
    },
    {
        "name": "Priya",
        "age": 19,
        "course": "IT",
        "marks": 90
    },
    {
        "name": "Amit",
        "age": 21,
        "course": "CSE",
        "marks": 78
    }
]
```

Output:



Index.html

```
<!DOCTYPE html>
```

```
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Student Hub Home</title>

    <link rel="stylesheet" href="../style.css">
    <link rel="stylesheet" href="../css/about.css">
    <link rel="stylesheet" href="../css/assignment.css">
    <link rel="stylesheet" href="../css/attendence.css">
    <link rel="stylesheet" href="../css/contact.css">
    <link rel="stylesheet" href="../css/course.css">
    <link rel="stylesheet" href="../css/login.css">
    <link rel="stylesheet" href="../css/profile.css">
    <link rel="stylesheet" href="../css/result.css">
    <link rel="stylesheet" href="../css/studentfeedback.css">

    <style>
        .data-section {
            margin: 25px 0;
            padding: 20px;
            border: 1px solid #ddd;
            border-radius: 10px;
            overflow-x: auto;
        }
        .data-section input, .data-section button {
            padding: 10px;
            margin: 5px;
        }
        .data-section button { cursor: pointer; }
        .data-table {
            width: 100%;
            border-collapse: collapse;
            margin-top: 15px;
        }
        .data-table th, .data-table td {
            border: 1px solid #ccc;
            padding: 10px;
            text-align: left;
        }
```

```
        .data-table th { background-color: #e8eef8; }
        #message, #studentMessage, #csvMessage {
            margin-top: 12px;
            font-weight: bold;
        }
    </style>
</head>
<body>
    <header>
        <h2>Student Hub</h2>
    </header>

    <img src="../images/charusat.png"
         alt="CHARUSAT University Logo" width="10%" height="60">

    <nav>
        <a href="index.html">Dashboard</a>
        <a href="login.html">Login</a>
        <a href="profile.html">Profile</a>
        <a href="about.html">About</a>
        <a href="assignment.html">Assignment</a>
        <a href="attendance.html">Attendance</a>
        <a href="contact.html">Contact</a>
        <a href="course.html">Course</a>
        <a href="result.html">Result</a>
        <a href="studentsfeedback.html">Student Feedback</a>
    </nav>

    <main>
        <h1>Welcome to Student Hub Portal</h1>
        <p>This portal provides information about assignments, attendance,
           contact, courses, login, results and student feedback.</p>
        <hr>

        <section class="data-section">
            <h2>Regex Validation Demo</h2>
            <form id="validationForm" novalidate>
                <label for="email">Email Address:</label><br>
                <input type="text" id="email" placeholder="Enter your
email" required>
```

```
                <br><br>
                <label for="password">Password:</label><br>
                <input type="password" id="password" placeholder="Enter
your password" required>
                <br><br>
                <button type="submit">Validate</button>
                <button type="reset">Reset</button>
                <p id="message" role="status" aria-live="polite"></p>
            </form>
        </section>

        <hr>

        <section class="data-section">
            <h2>Student Details (JSON Data)</h2>
            <p>Student information is loaded from students.json.</p>

            <input type="text" id="studentSearch" placeholder="Search by
student ID or name">
            <button type="button" id="searchStudent">Search
Student</button>
            <button type="button" id="showAllStudents">Show All
Students</button>
            <button type="button" id="loadStudents">Load JSON
Data</button>
            <p id="studentMessage" role="status">Click Load JSON Data to
begin.</p>

            <table class="data-table">
                <thead>
                    <tr>
                        <th>Student ID</th><th>Name</th><th>Email</th>
                        <th>Course</th><th>Semester</th>
                    </tr>
                </thead>
                <tbody id="studentTable">
                    <tr><td colspan="5">Click Load JSON Data to load
records.</td></tr>
                </tbody>
            </table>
```

```
        </section>

        <hr>

        <section class="data-section">
            <h2>Assignment Details (CSV Data)</h2>
            <p>Assignment records are loaded from assignments.csv.</p>
            <button type="button" id="loadAssignments">Load CSV
Data</button>
            <p id="csvMessage" role="status">Click Load CSV Data to
begin.</p>

            <table class="data-table">
                <thead>
                    <tr>
                        <th>Assignment
ID</th><th>Subject</th><th>Title</th>
                        <th>Due Date</th><th>Status</th>
                    </tr>
                </thead>
                <tbody id="assignmentTable">
                    <tr><td colspan="5">Click Load CSV Data to load
records.</td></tr>
                </tbody>
            </table>
        </section>
    </main>

    <footer>
        <h6>&copy; 2026 CHARUSAT University | Made by ID No. 25DCE023</h6>
    </footer>

    <script src="../js/about.js"></script>
    <script src="../js/assignment.js"></script>
    <script src="../js/attendence.js"></script>
    <script src="../js/contact.js"></script>
    <script src="../js/course.js"></script>
    <script src="../js/login.js"></script>
    <script src="../js/profile.js"></script>
    <script src="../js/result.js"></script>
```

```
    <script src="../js/studentfeedback.js"></script>

    <script>
        // 1. REGEX VALIDATION
        const emailRegex =
/^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/;
        const passwordRegex =
/^(?=.*[a-z])(?=.*[A-Z])(?=.*\d)(?=.*[@$!%*?&])[A-Za-z\d@$!%*?&]{8,}$/;

        const form = document.getElementById("validationForm");
        const message = document.getElementById("message");

        form.addEventListener("submit", function (event) {
            event.preventDefault();
            const email = document.getElementById("email").value.trim();
            const password = document.getElementById("password").value;

            if (email === "" || password === "") {
                message.textContent = "Please fill in all fields.";
            } else if (!emailRegex.test(email)) {
                message.textContent = "Please enter a valid email
address.";
            } else if (!passwordRegex.test(password)) {
                message.textContent = "Password must contain at least 8
characters, one uppercase, one lowercase, one number and one special
character (@$!%*?&).";
            } else {
                message.textContent = "Email and password format validated
successfully!";
            }
        });

        form.addEventListener("reset", function () {
            message.textContent = "";
        });

        // 2. LOAD STUDENTS FROM JSON
        let students = [];
        const studentTable = document.getElementById("studentTable");
        const studentMessage = document.getElementById("studentMessage");
```

```
        const studentSearch = document.getElementById("studentSearch");

        function displayStudents(records) {
            studentTable.replaceChildren();

            if (records.length === 0) {
                const row = studentTable.insertRow();
                const cell = row.insertCell();
                cell.colSpan = 5;
                cell.textContent = "No student records found.";
                return;
            }

            records.forEach(function (student) {
                const row = studentTable.insertRow();
                [student.id, student.name, student.email, student.course,
student.semester]
                    .forEach(function (value) {
                        const cell = row.insertCell();
                        cell.textContent = value;
                    });
            });
        }

        function loadStudents() {
            studentMessage.textContent = "Loading JSON data...";
            fetch("../data/students.json")
                .then(function (response) {
                    if (!response.ok) throw new Error("Could not load
students.json");
                    return response.json();
                })
                .then(function (data) {
                    if (!Array.isArray(data)) throw new Error("Invalid
JSON format");
                    students = data;
                    displayStudents(students);
                    studentMessage.textContent = "JSON loaded
successfully! Total students: " + students.length;
                })
```

```
                .catch(function () {
                    studentMessage.textContent = "Could not load JSON.
Check the data folder and use Live Server.";
                });
        }

        document.getElementById("loadStudents").addEventListener("click",
loadStudents);

        document.getElementById("searchStudent").addEventListener("click",
function () {
            if (students.length === 0) {
                studentMessage.textContent = "Please load JSON data
first.";
                return;
            }
            const query = studentSearch.value.trim().toLowerCase();
            const result = students.filter(function (student) {
                return String(student.id).toLowerCase().includes(query) ||
                       String(student.name).toLowerCase().includes(query);
            });
            displayStudents(result);
            studentMessage.textContent = "Matching records: " +
result.length;
        });


document.getElementById("showAllStudents").addEventListener("click",
function () {
            if (students.length === 0) {
                loadStudents();
                return;
            }
            studentSearch.value = "";
            displayStudents(students);
            studentMessage.textContent = "Showing all " + students.length
+ " students.";
        });

        // 3. LOAD ASSIGNMENTS FROM CSV
```

```
        const assignmentTable =
document.getElementById("assignmentTable");
        const csvMessage = document.getElementById("csvMessage");


document.getElementById("loadAssignments").addEventListener("click",
function () {
            csvMessage.textContent = "Loading CSV data...";
            fetch("../data/assignments.csv")
                .then(function (response) {
                    if (!response.ok) throw new Error("Could not load
assignments.csv");
                    return response.text();
                })
                .then(function (csvText) {
                    const lines = csvText.trim().split(/\r?\n/);
                    if (lines.length < 2) throw new Error("CSV has no
records");

                    const headers = lines[0].replace(/^\uFEFF/,
"").split(",");
                    const expected = ["assignmentId", "subject", "title",
"dueDate", "status"];
                    if (headers.length !== expected.length ||
                        !expected.every(function (header, index) { return
headers[index] === header; })) {
                        throw new Error("CSV headers do not match");
                    }

                    assignmentTable.replaceChildren();
                    lines.slice(1).forEach(function (line) {
                        const values = line.split(",");
                        if (values.length !== 5) return;
                        const row = assignmentTable.insertRow();
                        values.forEach(function (value) {
                            row.insertCell().textContent = value.trim();
                        });
                    });
                    csvMessage.textContent = "CSV loaded successfully!";
                })
```

```
                .catch(function () {
                    csvMessage.textContent = "Could not load CSV. Check
the data folder and use Live Server.";
                });
        });
    </script>
</body>
</html>
```

students.JSON

```
[
  {
    "id": "25DCE023",
    "name": "Rahul Patel",
    "email": "rahul@example.com",
    "course": "Computer Science Engineering",
    "semester": 3
  },
  {
    "id": "25DCE024",
    "name": "Priya Shah",
    "email": "priya@example.com",
    "course": "Computer Science Engineering",
    "semester": 3
  },
  {
    "id": "25DCE025",
    "name": "Amit Patel",
    "email": "amit@example.com",
    "course": "Information Technology",
    "semester": 3
  }
]
```

Assignment.csv

assignments.csv

```
assignmentId,subject,title,dueDate,status
A101,Java,OOP Concepts,2026-10-20,Pending
A102,DSA,Binary Tree,2026-10-22,Completed
A103,Web Development,HTML Forms,2026-10-25,Pending
A104,Computer Networks,IP Addressing,2026-10-28,Completed
```

Output

